

ONLINE LEARNING

Data Analyst Mindset Course

Discover premium online courses and grow your skills with expert instructors in data analysis, AI, business intelligence, and career development.



Mindset



Statistics



Excel



SQL



Power BI



Tableau



Python



AI Tools



Soft Skills



Projects

FOUNDATION

01

Introduction to Data Analysis

The Analyst's Mindset

- Problem Deconstruction & Logic
- Understanding the Data Life Cycle
- Business Context & Objective Alignment

Foundational Concepts

- Types of Data (Qualitative vs. Quantitative)
- Data Structures (Structured, Semi-structured, Unstructured)

Workflow & Strategy

- Defining Research Questions
- Data Collection & Governance Basics
- Interpreting Results vs. Reporting Numbers

FOUNDATION

02

Statistics for Data Analysis

Descriptive Statistics

- Mean, Median, Mode
- Range, Variance, Standard Deviation
- Data Distribution & Skewness

Inferential Statistics

- Population vs. Sample
- Confidence Intervals
- Hypothesis Testing Basics

TECHNICAL

03

Excel for
Data Analysis**Module 1: Basics & Interface**

- Introduction to Excel
- Workbook vs Worksheets
- Navigation & Shortcuts
- Working with Large Datasets

Module 2: Data Entry & Formatting

- Data Entry
- Cell Formatting
- Conditional Formatting
- Sorting & Filtering

Module 3: Formulas & Functions

- Arithmetic Operations
- SUM, AVERAGE, AutoFill
- Named Ranges
- Relative vs Absolute References

Module 4: Tables & Charts

- Tables & Charts
- Customization
- Error Handling

Module 5: Logical Functions

- IF, IFS, AND, OR
- Drop-down Lists

Module 6: Lookup & Text

- VLOOKUP, XLOOKUP
- INDEX & MATCH
- Text Functions

Module 7: Date Functions

- TODAY, NOW, DATEDIF

Module 8: Data Cleaning

- Remove Duplicates
- Text to Columns
- Advanced Filtering

Module 9: Pivot Tables

- Pivot Tables & Charts
- Calculated Fields

Module 10: AI & Tips

- AI in Excel
- Productivity Tips

Module 11: Final Project

- End-to-End Analysis Project

TECHNICAL

04

SQL for
Data Analysis**Module 1: Introduction**

- What is a Database
- Why Databases
- Relational Databases

Module 2: Modeling

- ERD Diagrams
- Relationships

Module 3: SQL Basics

- Syntax
- Data Types

Module 4: Core Operations

- SELECT
- WHERE
- ORDER BY

Module 5: Filtering

- AND, OR
- IN, BETWEEN
- LIKE

Module 6: Keys & Indexes

- Primary Key
- Foreign Key
- Indexes

Module 7: Joins

- Inner Join
- Outer Join
- Full Join
- Cross Join
- Self Join

Module 8: Aggregation

- COUNT, SUM, AVG
- GROUP BY
- HAVING

Module 9: Advanced

- Subqueries
- Views

Module 10: Window Functions

- ROW_NUMBER, RANK

Module 11: CTE

- Common Table Expressions

Module 12: Final Project

- SQL Analysis Project

TECHNICAL

05

Power BI

Module 1: Introduction

- Overview & Components
- Workflow & Interface

Module 2: Data Import

- Excel, CSV, SQL
- Online Sources

Module 3: Power Query

- Cleaning & Transformations
- Merging Queries

Module 4: Data Modeling

- Relationships
- Star Schema
- Optimization

Module 5: Visualization

- Charts & Dashboards
- Customization

Module 6: Interactivity

- Filters & Slicers
- Drillthrough
- Bookmarks

Module 7: DAX Basics

- SUM, COUNT
- Measures & Columns

Module 8: Advanced DAX

- IF, SWITCH
- Time Intelligence
- RANKX

Module 9: Sharing

- Publishing
- Exporting
- Optimization

Module 10: AI

- AI Visuals
- Insights

Module 11: Storytelling

- Dashboard Design
- Presenting Insights

Module 12: Final Project

- Dashboard Project

TECHNICAL

06

Tableau

Module 1: Introduction

- Overview & Components
- Workflow & Interface

Module 2: Data Connections

- File Sources (Excel, CSV, PDF)
- Server Sources (SQL, Cloud)
- Live vs. Extract
- Data Interpreter

Module 3: Data Preparation

- Relationships & Joins
- Data Blending
- Unions
- Pivoting & Splitting

Module 4: Calculations

- Calculated Fields
- Table Calculations
- LOD Expressions
- Sets & Parameters

Module 5: Dashboards & Stories

- Visualizations & Charts
- Dashboard Actions
- Formatting & Layout
- Story Points

TECHNICAL

07

Python for Data Analysis

Module 1: Introduction

- Programming Basics
- Python Overview
- Anaconda & Jupyter

Module 2: Basics

- Variables
- Data Types
- Strings
- Mutable vs Immutable

Module 3: Data Structures

- List
- Tuple
- Set
- Dictionary

Module 4: Conditions

- Comparison Operators
- Logical Operators
- If Statements

Module 5: Loops

- For Loop
- While Loop

Module 6: Functions

- Functions
- Parameters
- Return

Module 7: Libraries

- NumPy
- Pandas

Module 8: Visualization

- Matplotlib
- Seaborn
- Plotly

Module 9: Workflow

- Cleaning
- Analysis
- Visualization

Module 10: Automation

- Scripts
- File Handling

Module 11: AI

- AI in Python
- Use Cases

Module 12: Final Project

- Full Python Analysis Project

TECHNICAL

08

AI for Data Analysis

AI in Excel

- AI-Powered Formula Generation
- Natural Language Queries
- Automated Data Cleaning & Pattern Recognition

AI in Power BI

- Copilot for DAX & Report Creation
- AI Visuals (Key Influencers)
- Smart Narratives & Automated Summaries

AI in Python

- Generative AI for Pandas & Visualization
- Automated Script Debugging & Optimization
- Documentation & Comment Automation

AI-Driven Insights

- Automated Data Profiling
- Anomaly & Outlier Detection
- Automated Trend & Pattern Extraction
- Actionable Recommendations

SOFT SKILLS

09

Professional Development

Business Analysis

- Business Objectives & KPIs
- Problem Framing & Scoping
- Requirement Gathering
- Stakeholder Communication

Data Analyst Mindset

- Logical Thinking & Deconstruction
- Hypothesis Testing
- Critical Thinking in Analysis
- Actionable Insight Generation

Soft Skills

- Data Storytelling
- Presentation & Visualization Design
- Emotional Intelligence (EQ)
- Time Management & Planning

Freelance

- Platform Selection (Upwork, Freelancer)
- Profile Optimization
- Proposal Writing
- Project Pricing & Client Management

Portfolio & LinkedIn

- Project Documentation (GitHub/Medium)
- Personal Website / Portfolio Builder
- LinkedIn Profile Optimization
- Job Search & Interview Prep

PROJECTS

10

Domain-Based Projects

Real-World Case Studies

- Business & Marketing Analytics
- Medical & Healthcare Data
- Banking & Loans Risk Assessment
- Supply Chain & Logistics
- Sales & Revenue Performance